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Course outline

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- Lec 01 : Introduction and Recent Advances
- Lec 02 : Introduction to Natural Language Processing
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Week 1 : Assignment 1

The due date for submitting this assignment has passed.

Due on 2025-02-05, 23:59 IST.

As per our records you have not submitted this assignment.

1) Based on Distributional Semantics, which of the following statements is/are true? *1 point*

(i) The meaning of a word is defined by its relationship to other words.
(ii) The meaning of a word does not rely on its surrounding context.

- Both (i) and (ii) are correct
- Only (ii) is correct
- Only (i) is correct
- Neither (i) nor (ii) is correct

No, the answer is incorrect.
Score: 0

Accepted Answers:
Neither (i) nor (ii) is correct

2) Which of the following words have multiple senses? *1 point*

- light
- order
- letter
- buffalo

No, the answer is incorrect.
Score: 0

Accepted Answers:
light
order
letter
buffalo

3) Consider the following sentences: *1 point*

Sentence 1: Amit forgot to set an alarm last night.
Sentence 2: Amit woke up late today.

Does Sentence 1 entail Sentence 2?

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
False

4) What issues can be observed in the following text? *1 point*

On a much-needed #workcation in beautiful Goa. Workin & chillin by d waves!

- Idioms
- Non-standard English
- Tricky Entity Names
- Neologisms

No, the answer is incorrect.
Score: 0

Accepted Answers:
Non-standard English
Neologisms

5) Consider the following sentences: *1 point*

Sentence 1: The bats flew out of the cave at sunset.
Sentence 2: Rohan bought a new bat to practice cricket.

Question: Does the word "bat" have the same meaning in both sentences?

- Yes
- No

No, the answer is incorrect.
Score: 0

Accepted Answers:
No

6) Which of the following statements is/are true? *1 point*

- Apple is a hypernym of fruit.
- Leaf is a meronym of tree.
- Flower is a holonym of petal.
- Parrot is a hyponym of bird.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Leaf is a meronym of tree.
Flower is a holonym of petal.
Parrot is a hyponym of bird.

7) _____ deals with word formation and internal structure of words. *1 point*

- Pragmatics
- Discourse
- Semantics
- Morphology

No, the answer is incorrect.
Score: 0

Accepted Answers:
Morphology

8) Consider the following sentences: *1 point*

Sentence 1: Priya told Meera that she had completed the report on time.
Sentence 2: Meera was impressed by her dedication.

Which of the following statements is/are true?

- In Sentence 1, "she" refers to Meera.
- In Sentence 1, "she" refers to Priya.
- In Sentence 2, "her" refers to Priya.
- In Sentence 2, "her" refers to Meera.

No, the answer is incorrect.
Score: 0

Accepted Answers:
In Sentence 1, "she" refers to Priya.
In Sentence 2, "her" refers to Priya.

9) In semantic role labelling, we determine the semantic role of each argument with respect to the _____ of the sentence. *1 point*

- noun phrase
- subject
- predicate
- adjunct

No, the answer is incorrect.
Score: 0

Accepted Answers:
predicate

10) Which of the following statements is/are true? *1 point*

(i) Artificial Intelligence (AI) is a sub-field of Machine Learning.
(ii) LLMs are deep neural networks for processing text.
(iii) Generative AI (GenAI) involves only Large Language Models (LLMs).

- Only (i) and (ii) are correct
- Only (i) is correct
- Only (ii) and (iii) are correct
- All of (i), (ii), and (iii) are correct
- Neither (i), (ii), or (iii) is correct

No, the answer is incorrect.
Score: 0

Accepted Answers:
Only (ii) is correct

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- Lec 03 : Introduction to Statistical Language Models
- Lec 04 : Statistical LM: Advanced Smoothing and Evaluation
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Week 2 : Assignment 2

The due date for submitting this assignment has passed.

Due on 2025-02-05, 23:59 IST.

As per our records you have not submitted this assignment.

1) A 5-gram model is a _____ order Markov Model. *1 point*

- Constant
- Five
- Six
- Four

No, the answer is incorrect.
Score: 0
Accepted Answers:
Four

2) For a given corpus, the count of occurrence of the unigram "stay" is 300. If the Maximum Likelihood Estimation (MLE) for the bigram "stay curious" is 0.4, what is the count of occurrence of the bigram "stay curious"? *1 point*

- 123
- 300
- 750
- 120

No, the answer is incorrect.
Score: 0
Accepted Answers:
120

3) Which of the following are governing principles for Probabilistic Language Models? *1 point*

- Chain Rule of Probability
- Markov Assumption
- Fourier Transform
- Gradient Descent

No, the answer is incorrect.
Score: 0
Accepted Answers:
Chain Rule of Probability
Markov Assumption

For Question 4 to 5, consider the following corpus:

<s>the sunset is nice</s>
<s>people watch the sunset</s>
<s>they enjoy the beautiful sunset</s>

4) Assuming a bi-gram language model, calculate the probability of the sentence: *2 points*

<s>people watch the beautiful sunset</s>
Ignore the unigram probability of P(<s>) in your calculation.

- 2/27
- 1/27
- 2/9
- 1/6

No, the answer is incorrect.
Score: 0
Accepted Answers:
2/27

5) Assuming a bi-gram language model, calculate the perplexity of the sentence: *2 points*

<s>people watch the beautiful sunset</s>
Do not consider <s>and</s> in the count of words of the sentence.

- $27^{1/4}$
- $27^{1/5}$
- $9^{1/5}$
- $(27/2)^{1/5}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(27/2)^{1/5}$

6) What is the main intuition behind Kneser-Ney smoothing? *1 point*

- Assign higher probability to frequent words.
- Use continuation probability to better model words appearing in a novel context.
- Normalize probabilities by word length.
- Minimize perplexity for unseen words

No, the answer is incorrect.
Score: 0
Accepted Answers:
Use continuation probability to better model words appearing in a novel context.

7) In perplexity-based evaluation of a language model, what does a lower perplexity score indicate? *1 point*

- Worse model performance
- Better language model performance
- Increased vocabulary size
- More sparse data

No, the answer is incorrect.
Score: 0
Accepted Answers:
Better language model performance

8) Which of the following is a limitation of statistical language models like n-grams? *1 point*

- Fixed context size
- High memory requirements for large vocabularies
- Difficulty in generalizing to unseen data
- All of the above

No, the answer is incorrect.
Score: 0
Accepted Answers:
All of the above

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- Lec 06 : Introduction to PyTorch
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Week 3 : Assignment 3

The due date for submitting this assignment has passed.

Due on 2025-02-12, 23:59 IST.

As per our records you have not submitted this assignment.

1) State whether the following statement is True/False. *1 point*

The Perceptron learning algorithm can solve problems with non-linearly separable data.

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
False

2) In backpropagation, which method is used to compute the gradients? *1 point*

- Gradient descent
- Chain rule of derivatives
- Matrix factorization
- Linear regression

No, the answer is incorrect.
Score: 0

Accepted Answers:
Chain rule of derivatives

3) Which activation function outputs values in the range [−1,1]? *1 point*

- ReLU
- Tanh
- Sigmoid
- Linear

No, the answer is incorrect.
Score: 0

Accepted Answers:
Tanh

4) What is the primary goal of regularization in machine learning? *1 point*

- To improve the computational efficiency of the model
- To reduce overfitting
- To increase the number of layers in a network
- To minimize the loss function directly

No, the answer is incorrect.
Score: 0

Accepted Answers:
To reduce overfitting

5) Which of the following is a regularization technique where we randomly deactivate neurons during training? *1 point*

- Early stopping
- L1 regularization
- Dropout
- Weight decay

No, the answer is incorrect.
Score: 0

Accepted Answers:
Dropout

6) Which activation function has the vanishing gradient problem for large positive or negative inputs? *1 point*

- ReLU
- Sigmoid
- GELU
- Swish

No, the answer is incorrect.
Score: 0

Accepted Answers:
Sigmoid

7) Which activation function is defined as: $f(x)=x \cdot \sigma(x)$, where $\sigma(x)$ is the sigmoid function? *1 point*

- Swish
- ReLU
- GELU
- SwiGLU

No, the answer is incorrect.
Score: 0

Accepted Answers:
Swish

8) What does the backpropagation algorithm compute in a neural network? *1 point*

- Loss function value at each epoch
- Gradients of the loss function with respect to weights of the network
- Activation values of the output layer
- Output of each neuron

No, the answer is incorrect.
Score: 0

Accepted Answers:
Gradients of the loss function with respect to weights of the network

9) Which type of regularization encourages sparsity in the weights? *1 point*

- L1 regularization
- L2 regularization
- Dropout
- Early stopping

No, the answer is incorrect.
Score: 0

Accepted Answers:
L1 regularization

10) What is the main purpose of using hidden layers in an MLP? *1 point*

- Helps to the network bigger
- Enables us to handle linearly separable data
- Learn complex and nonlinear relationships in the data
- Minimize the computational complexity

No, the answer is incorrect.
Score: 0

Accepted Answers:
Learn complex and nonlinear relationships in the data

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- Lec 07 : Word Representation: Word2Vec & fastText
- Lec 08 : Word Representation: GloVe
- Lec 09 : Tokenization Strategies
- Lecture Material
- Feedback Form
- Quiz: Week 4 : Assignment 4

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Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2025-02-19, 23:59 IST.

As per our records you have not submitted this assignment.

1) What is the main drawback of representing words as one-hot vectors? *1 point*

- They cannot capture semantic similarity between words.
- They are computationally inefficient.
- They cannot incorporate word order effectively.
- They are not robust to unseen words.

No, the answer is incorrect.
Score: 0

Accepted Answers:
They cannot capture semantic similarity between words.

2) What is the key concept underlying Word2Vec? *1 point*

- Ontological semantics
- Decompositional semantics
- Distributional semantics
- Morphological analysis

No, the answer is incorrect.
Score: 0

Accepted Answers:
Distributional semantics

3) Why is sub-sampling frequent words beneficial in Word2Vec? *1 point*

- It increases the computational cost.
- It helps reduce the noise from high-frequency words.
- It helps eliminate redundancy.
- It prevents the model from learning embeddings for common words.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It helps reduce the noise from high-frequency words.

4) Which word relations cannot be captured by word2vec? *1 point*

- Polysemy
- Antonymy
- Analogy
- All of the these

No, the answer is incorrect.
Score: 0

Accepted Answers:
Polysemy
Antonymy

For Question 5 to 6, Consider the following word-word matrix:

	w_6	w_7	w_8	w_9	w_{10}	w_{11}	w_{12}
w_1	1	5	3	0	1	5	7
w_2	4	2	4	1	8	2	0
w_3	2	1	9	2	5	1	5
w_4	5	0	7	4	2	0	4
w_5	3	5	1	0	1	2	1

5) Compute the cosine similarity between w_2 and w_5. *1 point*

- 0.516
- 0.881
- 0.705
- 0.641

No, the answer is incorrect.
Score: 0

Accepted Answers:
0.641

6) Which word is most similar to w_1 based on cosine similarity? *4 points*

- w_2
- w_3
- w_4
- w_5

No, the answer is incorrect.
Score: 0

Accepted Answers:
w_5

7) What is the difference between CBOW and Skip-Gram in Word2Vec? *1 point*

- CBOW predicts the context word given the target word, while Skip-Gram predicts the target word given the context words.
- CBOW predicts the target word given the context words, while Skip-Gram predicts the context words given the target word.
- CBOW is used for generating word vectors, while Skip-Gram is not.
- Skip-Gram uses a thesaurus, while CBOW does not.

No, the answer is incorrect.
Score: 0

Accepted Answers:
CBOW predicts the target word given the context words, while Skip-Gram predicts the context words given the target word.

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- Lec 10 : Neural Language Models: CNN & RNN
- Lec 11 : Neural Language Models: LSTM & GRU
- Lec 12 : Sequence-to-Sequence Models
- Lec 13 : Decoding Strategies
- Lec 14 : Attention in Sequence-to-Sequence Models
- Lecture Material
- Feedback Form
- Quiz: Week 5 : Assignment 5

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Week 5 : Assignment 5

The due date for submitting this assignment has passed.

Due on 2025-02-26, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following is a disadvantage of Recurrent Neural Networks (RNNs)? ***1 point***

- Can only process fixed-length inputs.
- Symmetry in how inputs are processed.
- Difficulty accessing information from many steps back.
- Weights are not reused across timesteps.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Difficulty accessing information from many steps back.

2) Why are RNNs preferred over fixed-window neural models? ***1 point***

- They have a smaller parameter size.
- They can process sequences of arbitrary length.
- They eliminate the need for embedding layers.
- None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
They can process sequences of arbitrary length.

3) What is the primary purpose of the cell state in an LSTM? ***1 point***

- Store short-term information.
- Control the gradient flow across timesteps.
- Store long-term information.
- Perform the activation function.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Store long-term information.

4) In training an RNN, what technique is used to calculate gradients over multiple timesteps? ***1 point***

- Backpropagation through Time (BPTT)
- Stochastic Gradient Descent (SGD)
- Dropout Regularization
- Layer Normalization

No, the answer is incorrect.
Score: 0
Accepted Answers:
Backpropagation through Time (BPTT)

5) Consider a simple RNN: ***2 points***

- Input vector size: 3
- Hidden state size: 4
- Output vector size: 2
- Number of timesteps: 5

How many parameters are there in total, including the bias terms?

- 210
- 190
- 90
- 42

No, the answer is incorrect.
Score: 0
Accepted Answers:
42

6) What is the time complexity for processing a sequence of length 'N' by an RNN, if the input embedding dimension, hidden state dimension, and output vector dimension are all 'd'? ***1 point***

- O(N)
- $O(N^2d)$
- O(Nd)
- $O(Nd^2)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
O(Nd²)

7) Which of the following is true about Seq2Seq models? ***1 point***

(i) Seq2Seq models are always conditioned on the source sentence.
(ii) The encoder compresses the input sequence into a fixed-size vector representation.
(iii) Seq2Seq models cannot handle variable-length sequences.

- (i) and (ii)
- (ii) only
- (iii) only
- (i), (ii), and (iii)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(i) and (ii)

8) Given the following encoder and decoder hidden states, compute the attention scores. (Use dot product as the scoring function) ***2 points***

Encoder hidden states: $h_1 = [1,2]$, $h_2 = [3,4]$, $h_3 = [5,6]$
Decoder hidden state: $s = [0.5,1]$

- 0.00235,0.04731,0.9503
- 0.0737,0.287,0.6393
- 0.9503,0.0137,0.036
- 0.6393,0.0737,0.287

No, the answer is incorrect.
Score: 0
Accepted Answers:
0.00235,0.04731,0.9503

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How does an NPTEL online course work?

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- Lec 15 : Introduction to Transformer: Self & Multi-Head Attention
- Lec 16 : Introduction to Transformer: Positional Encoding and Layer Normalization
- Lec 17 : Implementation of Transformer using PyTorch
- Lecture Material
- Feedback Form
- Quiz: Week 6 : Assignment 6

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Week 6 : Assignment 6

The due date for submitting this assignment has passed.

Due on 2025-03-05, 23:59 IST.

As per our records you have not submitted this assignment.

1) What is the key advantage of multi-head attention? *1 point*

- It uses a single attention score for the entire sequence
- It allows attending to different parts of the input sequence simultaneously
- It eliminates the need for normalization
- It reduces the model size

No, the answer is incorrect.
Score: 0

Accepted Answers:
It allows attending to different parts of the input sequence simultaneously

2) What is the role of the residual connection in the Transformer architecture? *1 point*

- Improve gradient flow during backpropagation
- Normalize input embeddings
- Reduce computational complexity
- Prevent overfitting

No, the answer is incorrect.
Score: 0

Accepted Answers:
Improve gradient flow during backpropagation

3) Which of the following elements addresses the lack of sequence information in self-attention? *1 point*

- Non-linear transformations
- Positional encoding
- Masked decoding
- Residual connections

No, the answer is incorrect.
Score: 0

Accepted Answers:
Positional encoding

4) For Rotary Position Embedding (RoPE), which of the following statements are true? *1 point*

- Combines relative and absolute positional information
- Applies a multiplicative rotation matrix to encode positions
- Eliminates the need for positional encodings
- All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
Combines relative and absolute positional information
Applies a multiplicative rotation matrix to encode positions

5) Consider a sequence of tokens of length 4: $[w_1, w_2, w_3, w_4]$. Using masked self-attention, compute the attention weights for token w3, assuming the unmasked attention scores are: [5,2,1,3] *2 points*

- [0.6234, 0.023, 0.3424, 0.0112]
- [0.2957, 0.7043, 0, 0]
- [0.9362, 0.0466, 0.0171,0]
- [0.5061, 0.437, 0, 0.0569]

No, the answer is incorrect.
Score: 0

Accepted Answers:
[0.9362, 0.0466, 0.0171,0]

6) _____ maps the values of a feature in the range [0,1]. *1 point*

- Standardization
- Normalization
- Transformation
- Scaling

No, the answer is incorrect.
Score: 0

Accepted Answers:
Normalization

7) How does masked self-attention help in autoregressive models? *1 point*

- By attending to all tokens, including future ones.
- By focusing only on past tokens to prevent information leakage.
- By ignoring positional information in the sequence.
- By disabling the attention mechanism entirely.

No, the answer is incorrect.
Score: 0

Accepted Answers:
By focusing only on past tokens to prevent information leakage.

8) For a transformer with $d_{model} = 512$, calculate the positional encoding for position p=10 and dimensions 2 and 3 using the sinusoidal formula: *2 points*

$$PE(p,2i) = sin\left(\frac{p}{10000^{2i/d_{model}}}\right) \quad PE(p,2i+1) = cos\left(\frac{p}{10000^{2i/d_{model}}}\right)$$

- $sin\left(\frac{10}{10000^{1/256}}\right), cos\left(\frac{10}{10000^{1/256}}\right)$
- $cos\left(\frac{10}{10000^{1/512}}\right), sin\left(\frac{10}{10000^{1/512}}\right)$
- $cos\left(\frac{10}{10000^{4/512}}\right), sin\left(\frac{10}{10000^{7/256}}\right)$
- $sin\left(\frac{10}{10000^{2/512}}\right), cos\left(\frac{10}{10000^{3/512}}\right)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$sin\left(\frac{10}{10000^{1/256}}\right), cos\left(\frac{10}{10000^{1/256}}\right)$

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- Lec 18 : Pre-Training Strategies: ELMo, BERT
- Lec 19 : Pre-Training Strategies: Encoder-decoder and Decoder-only Models
- Lec 20 : Introduction to HuggingFace
- Lecture Material
- Feedback Form
- Quiz: Week 7 : Assignment 7

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Week 7 : Assignment 7

The due date for submitting this assignment has passed.

Due on 2025-03-12, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following best describes how ELMo's architecture captures different linguistic properties? *1 point*

- The model explicitly assigns specific linguistic functions to each layer.
- The lower layers capture syntactic information, while higher layers capture semantic information.
- All layers capture the similar properties.
- ELMo uses a fixed, non-trainable weighting scheme for combining layer-wise representations.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The lower layers capture syntactic information, while higher layers capture semantic information.

2) BERT and BART models differ in their architectures. While BERT is (i)-------- model, BART is (ii) --------- one. Select the correct choices for (i) and (ii). *1 point*

- i: Decoder-only , ii: Encoder-only
- i: Encoder-decoder , ii: Encoder-only
- i: Encoder-only , ii: Encoder-decoder
- i: Decoder-only , ii: Encoder-decoder

No, the answer is incorrect.
Score: 0
Accepted Answers:
i: Encoder-only , ii: Encoder-decoder

3) The pre-training objective for the T5 model is based on: *1 point*

- Next sentence prediction
- Masked language modelling
- Span corruption and reconstruction
- Predicting the next token

No, the answer is incorrect.
Score: 0
Accepted Answers:
Span corruption and reconstruction

4) Which of the following datasets was used to pretrain the T5 model? *1 point*

- Wikipedia
- BookCorpus
- Common Crawl
- C4

No, the answer is incorrect.
Score: 0
Accepted Answers:
C4

5) Which of the following special tokens are introduced in BERT to handle sentence pairs? *1 point*

- [MASK] and [CLS]
- [SEP] and [CLS]
- [CLS] and [NEXT]
- [SEP] and [MASK]

No, the answer is incorrect.
Score: 0
Accepted Answers:
[SEP] and [CLS]

6) ELMo and BERT represent two different pre-training strategies for language models. Which of the following statement(s) about these approaches is/are true? *2 points*

- [ ] ELMo uses a bi-directional LSTM to pre-train word representations, while BERT uses a transformer encoder with masked language modeling.
- [ ] ELMo provides context-independent word representations, whereas BERT provides context-dependent representations.
- [ ] Pre-training of both ELMo and BERT involve next token prediction.
- [ ] Both ELMo and BERT produce word embeddings that can be fine-tuned for downstream tasks.

No, the answer is incorrect.
Score: 0
Accepted Answers:
ELMo uses a bi-directional LSTM to pre-train word representations, while BERT uses a transformer encoder with masked language modeling.
Both ELMo and BERT produce word embeddings that can be fine-tuned for downstream tasks.

7) Decoder-only models are essentially trained based on probabilistic language modelling. Which of the following correctly represents the training objective of GPT-style models? *1 point*

- P(y | x) where x is the input sequence and y is the gold output sequence
- P(x | y) where x is the input sequence and y is the gold output sequence
- $P(w_t \mid w_{1:t-1})$, where w_t represents the token at position t, and $w_{1:t-1}$ is the sequence of tokens from position 1 to t-1
- $P(w_t \mid w_{1:t+1})$, where wt represents the token at position t, and $w_{1:t+1}$ is the sequence of tokens from position 1 to t+1

No, the answer is incorrect.
Score: 0
Accepted Answers:
$P(w_t \mid w_{1:t-1})$, where w_t represents the token at position t, and $w_{1:t-1}$ is the sequence of tokens from position 1 to t-1

8) In the previous week, we saw the usage of **einsum** function in numpy as a generalized operation for performing tensor multiplications. Now, consider two matrices: $A = \begin{bmatrix} 1 & 5 \\ 3 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 4 & 2 \end{bmatrix}$.

Then, what is the output of the following numpy operation?

```
numpy.einsum('ij, ij -> ', A, B)
```

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 23

2 points

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- Lec 21 : Instruction Tuning
- Lec 22 : Prompt-based Learning
- Lec 23 : Advanced Prompting and Prompt Sensitivity
- Lec 24 : Alignment of Language Models-I
- Lec 25 : Alignment of Language Models-II
- Lecture Material
- Feedback Form

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Week 8 : Assignment 8

The due date for submitting this assignment has passed.

Due on 2025-03-19, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which factors influence the effectiveness of instruction tuning? *1 point*

- [ ] The number of instruction templates used in training.
- [ ] The tokenization algorithm used by the model.
- [ ] The diversity of tasks in the fine-tuning dataset.
- [ ] The order in which tasks are presented during fine-tuning.

No, the answer is incorrect.
Score: 0

Accepted Answers:
The number of instruction templates used in training.
The diversity of tasks in the fine-tuning dataset.
The order in which tasks are presented during fine-tuning.

2) What are key challenges of soft prompts in prompt-based learning? *1 point*

- [ ] Forward pass with them is computationally inefficient compared to that with hard prompts.
- [ ] They require additional training, unlike discrete prompts.
- [ ] They cannot be interpreted or used effectively by non-expert users.
- [ ] They require specialized architectures that differ from standard transformers.

No, the answer is incorrect.
Score: 0

Accepted Answers:
They require additional training, unlike discrete prompts.
They cannot be interpreted or used effectively by non-expert users.

3) Which statement best describes the impact of fine-tuning versus prompting in LLMs? *1 point*

- ( ) Fine-tuning is always superior to prompting in generalization tasks.
- ( ) Prompting requires gradient updates, while fine-tuning does not.
- ( ) Fine-tuning modifies the model weights permanently, while prompting does not.
- ( ) Prompting performs better on in-domain tasks compared to fine-tuning.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Fine-tuning modifies the model weights permanently, while prompting does not.

4) Which of the following aspects of the model outputs are captured by POSIX? *1 point*

- [ ] Diversity in the responses to intent-preserving prompt variations
- [ ] Entropy of the distribution of response frequencies
- [ ] Time required to generate responses for intent-preserving prompt variations
- [ ] Variance in the log-likelihood of the same response for different input prompt variations

No, the answer is incorrect.
Score: 0

Accepted Answers:
Diversity in the responses to intent-preserving prompt variations
Entropy of the distribution of response frequencies
Variance in the log-likelihood of the same response for different input prompt variations

5) Which key mechanism makes Tree-of-Thought (ToT) prompting more effective than Chain-of-Thought (CoT)? *1 point*

- ( ) ToT uses reinforcement learning for better generalization.
- ( ) ToT allows backtracking to explore multiple reasoning paths.
- ( ) ToT reduces hallucination by using domain-specific heuristics.
- ( ) ToT eliminates the need for manual prompt engineering.

No, the answer is incorrect.
Score: 0

Accepted Answers:
ToT allows backtracking to explore multiple reasoning paths.

6) What is a key limitation of measuring accuracy alone when evaluating LLMs? *1 point*

- ( ) Accuracy is always correlated with model size.
- ( ) Accuracy cannot be measured on open-ended tasks.
- ( ) Accuracy is independent of the training dataset size.
- ( ) Accuracy does not account for prompt sensitivity.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Accuracy does not account for prompt sensitivity.

7) Why is instruction tuning not sufficient for aligning large language models? *1 point*

- ( ) It does not generalize to unseen tasks.
- ( ) It cannot prevent models from generating undesired responses.
- ( ) It reduces model performance on downstream tasks.
- ( ) It makes models less capable of learning from new data.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It cannot prevent models from generating undesired responses.

8) Why is KL divergence minimized in regularized reward maximization? *1 point*

- ( ) To maximize the probability of generating high-reward responses.
- ( ) To make training more computationally efficient.
- ( ) To prevent the amplification of bias in training data.
- ( ) To ensure models do not diverge too far from the reference model.

No, the answer is incorrect.
Score: 0

Accepted Answers:
To ensure models do not diverge too far from the reference model.

9) What is the primary advantage of using the log-derivative trick in REINFORCE? *1 point*

- ( ) Reducing data requirements
- ( ) Expanding the token vocabulary
- ( ) Simplifying gradient computation
- ( ) Improving sampling diversity

No, the answer is incorrect.
Score: 0

Accepted Answers:
Simplifying gradient computation

10) Which method combines reward maximization and minimizing KL divergence? *1 point*

- ( ) REINFORCE
- ( ) Monte Carlo Approximation
- ( ) Proximal Policy Optimization
- ( ) Constitutional AI

No, the answer is incorrect.
Score: 0

Accepted Answers:
Proximal Policy Optimization

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- Lec 26 : Knowledge and Retrieval: Knowledge Graph
- Lec 27 : Knowledge and Retrieval: Knowledge Graph Completion and Evaluation
- Lec 28 : Knowledge and Retrieval: Translation and Rotation Models
- Lecture Material
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Week 9 : Assignment 9

The due date for submitting this assignment has passed.

Due on 2025-03-26, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following statement best describes why knowledge graphs (KGs) are considered more powerful than a traditional relational knowledge base (KB)? *1 point*

- KGs require no schema, whereas KBs must have strict schemas.
- KGs store data only in the form of hypergraphs, eliminating redundancy.
- KGs allow flexible, graph-based connections and typed edges, enabling richer relationships and inferences compared to KBs.
- KGs completely replace the need for textual sources by storing all possible facts.

No, the answer is incorrect.
Score: 0
Accepted Answers:
KGs allow flexible, graph-based connections and typed edges, enabling richer relationships and inferences compared to KBs.

2) Entity alignment and relation alignment are crucial between KGs of different languages. Which of the following factors contribute to effective alignment? *1 point*

- Aligning relations solely by their lexical similarity, ignoring semantic context
- Transliteration or language-based string matching for entity labels
- Ensuring all language aliases are represented identically in each KG
- Matching neighbours, or connected entities, across different KGs

No, the answer is incorrect.
Score: 0
Accepted Answers:
Transliteration or language-based string matching for entity labels
Matching neighbours, or connected entities, across different KGs

3) In the context of knowledge graph completion (KGC), which statement best describes the role of the scoring function f(s,r,o)? *1 point*

- It determines whether two entities refer to the same real-world concept.
- It produces a raw confidence score indicating how plausible a triple (s,r,o) is.
- It explicitly encodes only the subject's embedding, ignoring the relation and object embeddings.
- It ensures that every negative triple gets a higher score than any positive triple.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It produces a raw confidence score indicating how plausible a triple (s,r,o) is.

4) One key difference between the differentiable KG approach and the semantic interpretation approach to KGQA is: *1 point*

- Differentiable KG approaches are fully rule-based, while semantic interpretation is purely neural.
- Differentiable KG approaches do not require any graph embeddings, relying instead on explicit logical forms.
- Semantic interpretation is more transparent or interpretable, whereas differentiable KG is end-to-end trainable but less interpretable.
- Both approaches use logical forms; the primary difference is the type of question they can answer.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Semantic interpretation is more transparent or interpretable, whereas differentiable KG is end-to-end trainable but less interpretable.

5) Considering the differentiable KG approach, which elements are typically learned jointly when training an end-to-end KGQA model? *1 point*

- The textual question representation (e.g., BERT embeddings)
- The graph structure encoding (e.g., GCN or transformer-based graph embeddings)
- Predefined logical forms to ensure interpretability
- The final answer selection mechanism that identifies which node(s) in the graph satisfy the question

No, the answer is incorrect.
Score: 0
Accepted Answers:
The textual question representation (e.g., BERT embeddings)
The graph structure encoding (e.g., GCN or transformer-based graph embeddings)
The final answer selection mechanism that identifies which node(s) in the graph satisfy the question

6) Uniform negative sampling can have high variance and may require large number of samples. Why is that the case? *1 point*

- Because the margin-based loss cannot converge without big mini-batches.
- Because randomly picking negative entities does not guarantee close or challenging negatives, causing unstable training estimates.
- Because negative sampling must ensure every possible negative triple is covered.
- Because the number of relations in the KG is too large for small number of samples.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Because randomly picking negative entities does not guarantee close or challenging negatives, causing unstable training estimates.

7) In testing embedding and score quality for KG completion, mean rank and hits@K are typical metrics. What does hits@K specifically measure in this context? *1 point*

- The percentage of queries for which the correct answer appears in the top-K of the ranked list.
- The reciprocal of the rank of the correct answer.
- The probability of the correct answer appearing as the highest scored candidate.
- The margin of the correct triple score relative to all negative triples.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The percentage of queries for which the correct answer appears in the top-K of the ranked list.

8) In the TransE model, the scoring function for a triple (s,r,o) is typically defined as *1 point*

$f(s,r,o)=\|e_s + e_r - e_o\|$

where e_s, e_r, e_o are embeddings of the subject, relation, and object, respectively. Which statement best explains what a low value of f(s,r,o) indicates in this context?

- That (s,r,o) is an invalid triple according to the learned embeddings.
- That e_s and e_o must be orthogonal.
- That the relation embedding e_r is zero.
- That (s,r,o) has a high likelihood of being a true fact in the knowledge graph.

No, the answer is incorrect.
Score: 0
Accepted Answers:
That (s,r,o) has a high likelihood of being a true fact in the knowledge graph.

9) In RotatE, if a relation r is intended to be symmetric, how would that typically manifest in the complex plane? *1 point*

- The relation embedding e_r must always equal zero.
- The angle of e_r must be $\pi/2$.
- The relation embedding e_r is its own inverse (i.e., a 180° rotation when squared).
- The magnitude of e_r must be greater than 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The relation embedding e_r is its own inverse (i.e., a 180° rotation when squared).

10) Which main advantage do rotation-based models (like RotatE) have over translation-based ones (like TransE) when it comes to complex multi-relational patterns in a KG? *1 point*

- Rotation-based models cannot model any symmetry or inverse patterns, so they are simpler.
- Rotation-based models handle a broader set of relation properties (symmetry, anti-symmetry, inverses, composition) more naturally.
- Rotation-based models have no hyperparameters to tune, unlike TransE.
- Rotation-based models are guaranteed to yield perfect link prediction.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Rotation-based models handle a broader set of relation properties (symmetry, anti-symmetry, inverses, composition) more naturally.

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- Lec 30 : Quantization, Pruning & Distillation
- Lec 31 : An Alternate Formulation of Transformers: Residual Stream Perspective
- Lec 32 : Interpretability Techniques
- Lecture Material
- Feedback Form

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Week 10 : Assignment 10

The due date for submitting this assignment has passed.

Due on 2025-04-02, 23:59 IST.

As per our records you have not submitted this assignment.

1) How do Prefix Tuning and Adapters differ in terms of where they inject new task-specific parameters in the Transformer architecture? *1 point*

- Prefix Tuning adds new feed-forward networks after every attention block, while Adapters prepend tokens.
- Both approaches modify only the final output layer but in different ways.
- Prefix Tuning learns trainable "prefix" hidden states at each layer's input, whereas Adapters insert small bottleneck modules inside the Transformer blocks.
- Both approaches rely entirely on attention masks to inject new task-specific knowledge.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Prefix Tuning learns trainable "prefix" hidden states at each layer's input, whereas Adapters insert small bottleneck modules inside the Transformer blocks.

2) The Structure-Aware Intrinsic Dimension (SAID) improves over earlier low-rank adaptation approaches by: *1 point*

- Ignoring the network structure entirely
- Learning one scalar per layer for layer-wise scaling
- Sharing the same random matrix across all layers
- Using adapters within self-attention layers

No, the answer is incorrect.
Score: 0

Accepted Answers:
Learning one scalar per layer for layer-wise scaling

3) Which of the following are correct about the extensions of LoRA? *1 point*

- LongLoRA supports inference on longer sequences using global attention
- QLoRA supports low-rank adaptation on 4-bit quantized models
- DyLoRA automatically selects the optimal rank during training
- LoRA+ introduces gradient clipping to stabilize training

No, the answer is incorrect.
Score: 0

Accepted Answers:
QLoRA supports low-rank adaptation on 4-bit quantized models
DyLoRA automatically selects the optimal rank during training

4) Which pruning technique specifically removes weights with the smallest absolute values first, potentially followed by retraining to recover accuracy? *1 point*

- Magnitude Pruning
- Structured Pruning
- Random Pruning
- Knowledge Distillation

No, the answer is incorrect.
Score: 0

Accepted Answers:
Magnitude Pruning

5) In Post-Training Quantization (PTQ) for LLMs, why is a calibration dataset used? *1 point*

- To precompute the entire attention matrix for all tokens.
- To remove outlier dimensions before applying magnitude-based pruning.
- To fine-tune the entire model on a small dataset and store the new weights.
- To estimate scale factors for quantizing weights and activations under representative data conditions.

No, the answer is incorrect.
Score: 0

Accepted Answers:
To estimate scale factors for quantizing weights and activations under representative data conditions.

6) Which best summarizes the function of the unembedding matrix W_U? *1 point*

- It merges the queries and keys for each token before final classification.
- It converts the final residual vector into vocabulary logits for next-token prediction.
- It is used for normalizing the QK and OV circuits so that their norms match.
- It acts as a second attention layer that aggregates multiple heads.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It converts the final residual vector into vocabulary logits for next-token prediction.

7) Which definition best matches an induction head as discovered in certain Transformer circuits? *1 point*

- A head that specifically attends to punctuation tokens to determine sentence boundaries
- A feed-forward sub-layer specialized for outputting next-token probabilities for out-of-distribution tokens
- A head that looks for previous occurrences of a token A, retrieves the token B that followed it last time, and then predicts B again
- A masking head that prevents the model from looking ahead at future tokens

No, the answer is incorrect.
Score: 0

Accepted Answers:
A head that looks for previous occurrences of a token A, retrieves the token B that followed it last time, and then predicts B again

8) In mechanistic interpretability, how can we define 'circuit'? *1 point*

- A data pipeline for collecting training examples in an autoregressive model
- A small LSTM module inserted into a Transformer for additional memory
- A device external to the neural network used to fine-tune certain parameters after training
- A subgraph of the neural network hypothesized to implement a specific function or behaviour

No, the answer is incorrect.
Score: 0

Accepted Answers:
A subgraph of the neural network hypothesized to implement a specific function or behaviour

9) Which best describes the role of Double Quantization in QLoRA? *1 point*

- It quantizes the attention weights twice to achieve 1-bit representations.
- It reinitializes parts of the model with random bit patterns for improved regularization.
- It quantizes the quantization constants themselves for additional memory savings.
- It systematically reverts partial quantized weights back to FP16 whenever performance degrades.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It quantizes the quantization constants themselves for additional memory savings.

10) Which of the following are true about sequence-level distillation for LLMs? *1 point*

- It trains a student model by matching the teacher's sequence outputs (e.g., predicted token sequences) rather than just individual token distributions.
- It requires storing only the top-1 predictions from the teacher model for each token.
- It can be combined with word-level distillation to transfer both local and global knowledge.
- It forces the teacher to produce a chain-of-thought explanation for each example.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It trains a student model by matching the teacher's sequence outputs (e.g., predicted token sequences) rather than just individual token distributions.
It can be combined with word-level distillation to transfer both local and global knowledge.

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- Lec 34 : Knowledge and Retrieval: Modeling Hierarchies
- Lec 35 : Knowledge and Retrieval: Temporal Knowledge Graphs
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Week 11 : Assignment 11

The due date for submitting this assignment has passed.

Due on 2025-04-09, 23:59 IST.

As per our records you have not submitted this assignment.

1) What is the main modification that SimplE makes to DistMult-like models to handle asymmetric relations? *1 point*

- Replacing entity embeddings with random fixed vectors
- Introducing separate entity embeddings for subject and object roles, along with inverse relations
- Restricting the rank of the relation tensor to 1
- Using negative sampling for half of the triple set

No, the answer is incorrect.
Score: 0

Accepted Answers:
Introducing separate entity embeddings for subject and object roles, along with inverse relations

2) Which statements correctly characterize the basic DistMult approach for knowledge graph completion? *1 point*

- Each relation r is parameterized by a full D×D matrix that can capture asymmetric relations.
- The relation embedding is a diagonal matrix, leading to a multiplicative interaction of entity embeddings.
- DistMult struggles with non-symmetric relations because score(s, r, o) = $a_s^T M_r a_o$ is inherently symmetric in s and o.
- DistMult's performance is typically tested only on fully symmetric KGs.

No, the answer is incorrect.
Score: 0

Accepted Answers:
The relation embedding is a diagonal matrix, leading to a multiplicative interaction of entity embeddings.
DistMult struggles with non-symmetric relations because score(s, r, o) = $a_s^T M_r a_o$ is inherently symmetric in s and o.

3) Which statements about the ComplEx extension of DistMult are true? *1 point*

- It uses complex-valued embeddings to better capture asymmetric or anti-symmetric relations.
- It replaces the multiplication in DistMult with element-wise addition of real-valued vectors.
- For a perfectly symmetric relation, one could set the imaginary part of the relation embedding to zero.
- ComplEx requires each entity vector to be unit norm in the complex plane.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It uses complex-valued embeddings to better capture asymmetric or anti-symmetric relations.
For a perfectly symmetric relation, one could set the imaginary part of the relation embedding to zero.

4) Which best describes the main advantage of using a factorized representation (e.g., DistMult, ComplEx) for large KGs? *1 point*

- It enforces that every relation in the KG be perfectly symmetric.
- It ensures each entity is stored as a one-hot vector, simplifying nearest-neighbour queries.
- It collapses the entire KG into a single scalar value.
- It significantly reduces parameters and enables generalization to unseen triples by capturing low-rank structure.

No, the answer is incorrect.
Score: 0

Accepted Answers:
It significantly reduces parameters and enables generalization to unseen triples by capturing low-rank structure.

5) Which statement best describes the reshaping of a 3D KG tensor $X \in \mathbb{R}^{|E|\times|R|\times|E|}$ into a matrix factorization problem? *1 point*

- One axis remains for subject, one axis remains for object, and relations are combined into a single expanded axis.
- The subject dimension is repeated to match the relation dimension, resulting in a 2D matrix.
- Each subject–relation pair is collapsed into a single dimension, while objects remain as separate entries.
- The entire KG is vectorized into a 1D array and then factorized with an SVD approach.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Each subject–relation pair is collapsed into a single dimension, while objects remain as separate entries.

6) Which key property of hierarchical relationships (e.g. is-a, transitivity) motivates the exploration of specialized embedding methods over standard Euclidean KG embeddings? *1 point*

- Symmetry in the relation (A, is-a, B) implying (B, is-a, A)
- Frequent presence of cycles in hierarchical graphs
- Transitivity in the form (camel, is-a, mammal) and (mammal, is-a, animal) ⇒ (camel,is-a,animal)
- The high dimensionality of the entity embeddings

No, the answer is incorrect.
Score: 0

Accepted Answers:
Transitivity in the form (camel, is-a, mammal) and (mammal, is-a, animal) ⇒ (camel,is-a,animal)

7) Which of the following statements correctly describe hyperbolic (Poincare) embeddings for hierarchical data? *1 point*

- They map nodes onto a disk (or ball) such that large branching factors can be represented with lower distortion than in Euclidean space.
- Distance grows slowly near the center and becomes infinite near the boundary, making it naturally suited for tree-like structures.
- They require each node to be embedded on the surface of the Poincare disk of radius 1.
- They can achieve arbitrarily low distortion embeddings for trees with the same dimension as Euclidean space.

No, the answer is incorrect.
Score: 0

Accepted Answers:
They map nodes onto a disk (or ball) such that large branching factors can be represented with lower distortion than in Euclidean space.
Distance grows slowly near the center and becomes infinite near the boundary, making it naturally suited for tree-like structures.

8) Why might a partial-order-based approach (like order embeddings) be beneficial for modelling 'is-a' relationships compared to purely distance-based approaches? *1 point*

- They explicitly encode the ancestor–descendant relation as a coordinate-wise inequality or containment.
- They can represent negative correlations (i.e., sibling vs. ancestor) more easily than distance metrics.
- They inherently guarantee transitive closure of the hierarchy in the learned embedding space.
- They do not rely on pairwise distances but use a notion of coordinate-wise ordering or interval containment.

No, the answer is incorrect.
Score: 0

Accepted Answers:
They explicitly encode the ancestor–descendant relation as a coordinate-wise inequality or containment.
They do not rely on pairwise distances but use a notion of coordinate-wise ordering or interval containment.

9) Which statement about box embeddings in hierarchical modelling is most accurate? *1 point*

- Each entity or type is assigned a single real-valued vector, ignoring bounding volumes.
- Containment $I_x \subset I_y$ across all dimensions encodes x<y .
- They rely on spherical distances around a central node to measure tree depth.
- They cannot be used to represent set intersections or partial overlap.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Containment $I_x \subset I_y$ across all dimensions encodes x<y .

10) What is a key challenge with axis-aligned open-cone (order) embeddings for hierarchical KG data? *1 point*

- They enforce that all sibling categories have identical cone apices, which causes overlap.
- They require symmetrical relationships for all edges.
- They do not allow partial orders to be extended to total orders.
- The volume (measure) of cones is the same regardless of how "broad" or "narrow" the cone is, making sub-categories indistinguishable by volume.

No, the answer is incorrect.
Score: 0

Accepted Answers:
The volume (measure) of cones is the same regardless of how "broad" or "narrow" the cone is, making sub-categories indistinguishable by volume.

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- Lec 37 : Conclusion: Expert Panel Discussion
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Week 12 : Assignment 12

The due date for submitting this assignment has passed.

Due on 2025-04-16, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which statements correctly characterize "bias" in the context of LLMs? *1 point*

1. Bias can generate objectionable or stereotypical views in model outputs.
2. Bias is always intentionally introduced by malicious data curators.
3. Bias can cause harmful real-world impacts such as reinforcing discrimination.
4. Bias only affects low-resource languages; high-resource languages are unaffected.

- 1 and 2
- 1 and 3
- 2 and 4
- 1, 3, and 4

No, the answer is incorrect.
Score: 0
Accepted Answers:
1 and 3

2) The Stereotype Score (ss) refers to: *1 point*

- The frequency with which a language model rejects biased associations.
- The measure of how often a model's predictions are meaningless as opposed to meaningful.
- A ratio of positive sentiment to negative sentiment in model outputs.
- The proportion of examples in which a model chooses a stereotypical association over an anti-stereotypical one.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The proportion of examples in which a model chooses a stereotypical association over an anti-stereotypical one.

3) Which of the following are prominent sources of bias in LLMs? *1 point*

1. Improper selection of training data leading to skewed distributions.
2. Reliance on older datasets causing "temporal bias."
3. Overemphasis on low-resource languages causing "linguistic inversion."
4. Unequal focus on high-resource languages resulting in "cultural bias."

- 1 and 2 only
- 2 and 3 only
- 1, 2, and 4
- 1, 3, and 4

No, the answer is incorrect.
Score: 0
Accepted Answers:
1, 2, and 4

4) In the context of bias mitigation based on adversarial triggers, which best describes the goal of prepending specially chosen tokens to prompts? *1 point*

- To directly fine-tune the model parameters to remove bias
- To override all prior knowledge in a model, effectively "resetting" it
- To exploit the model's distributional patterns, thereby neutralizing or flipping biased associations in generated text
- To randomly shuffle the tokens so that the model becomes more robust

No, the answer is incorrect.
Score: 0
Accepted Answers:
To exploit the model's distributional patterns, thereby neutralizing or flipping biased associations in generated text

5) Which of the following best describes the "regard" metric? *1 point*

- It is a measure of how well a model can explain its internal decision process.
- It is a measurement of a model's perplexity on demographically sensitive text.
- It is the proportion of times a model self-corrects discriminatory language.
- It is a classification label reflecting the attitude towards a demographic group in the generated text.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It is a classification label reflecting the attitude towards a demographic group in the generated text.

6) Which of the following steps compose the approach for improving response safety via in-context learning? *1 point*

- Retrieving safety demonstrations similar to the user query.
- Fine-tuning the model with additional labeled data after generation.
- Providing retrieved demonstrations as examples in the prompt to guide the model's response generation.
- Sampling multiple outputs from LLMs and choosing the majority opinion.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Retrieving safety demonstrations similar to the user query.
Providing retrieved demonstrations as examples in the prompt to guide the model's response generation.

7) Which statement(s) is/are correct about how high-resource (HRL) vs. low-resource languages (LRL) affect model training? *1 point*

- LRLs typically have higher performance metrics due to smaller population sizes.
- HRLs get more data, so the model might overfit to HRL cultural perspectives.
- LRLs are often under-represented, leading to potential underestimation of their cultural nuances.
- The dominance of HRLs can cause a reinforcing cycle that perpetuates imbalance.

No, the answer is incorrect.
Score: 0
Accepted Answers:
HRLs get more data, so the model might overfit to HRL cultural perspectives.
LRLs are often under-represented, leading to potential underestimation of their cultural nuances.
The dominance of HRLs can cause a reinforcing cycle that perpetuates imbalance.

8) The "Responsible LLM" concept is stated to address: *1 point*

- Only the bias in LLMs
- A set of concerns including explainability, fairness, robustness, and security
- Balancing training costs with carbon footprint
- Implementation of purely rule-based safety filters

No, the answer is incorrect.
Score: 0
Accepted Answers:
A set of concerns including explainability, fairness, robustness, and security

9) Within the StereoSet framework, the icat metric specifically refers to: *1 point*

- The ratio of anti-stereotypical associations to neutral associations
- The percentage of times a model refuses to generate content deemed hateful
- A measure of domain coverage across different demographic groups
- A balanced metric capturing both a model's language modelling ability and the tendency to avoid stereotypical bias

No, the answer is incorrect.
Score: 0
Accepted Answers:
A balanced metric capturing both a model's language modelling ability and the tendency to avoid stereotypical bias

10) Bias due to improper selection of training data typically arises in LLMs when: *1 point*

- Data are selected exclusively from curated, balanced sources with equal representation
- The language model sees only real-time social media feeds without any historical texts
- The training corpus over-represents some topics or groups, creating a skewed distribution
- All data are automatically filtered to remove any demographic markers

No, the answer is incorrect.
Score: 0
Accepted Answers:
The training corpus over-represents some topics or groups, creating a skewed distribution